Supplemental Material: Orthogonal mode analysis of two-dimensional small-angle neutron scattering

SANS2D mode analyzer contributors

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Abstract

This document describes the analysis implemented in `sans2d-mode-analyzer` (version 0.1.0). A real-valued detector pattern on a circular annulus is represented by orthonormal radial Legendre modes and real angular Fourier modes. We specify the linear fit, its covariance, the associated Parseval powers, angular block statistics and optional numerical Holm diagnostics, and the radial coefficient spectra. We also describe detector selection, optional pixel integration and parity-based centering, symmetry diagnostics, reduced reconstructions, truncation checks, and angular-template projections. All quantities refer to the chosen annulus and finite fitted expansion.

S1. METHODOLOGY OF THE TWO-DIMENSIONAL SANS MODE ANALYSIS

A. Scope, detector coordinates, and integration measure

The input is a reduced, real-valued scattering intensity or differential cross section, denoted by $I(q, \theta)$. The analysis describes its radial and angular dependence without prescribing a particle form factor or a magnetization model. In particular, the modal fit does not estimate vortex amplitudes, particle radii, or structure-factor parameters. Such physical interpretations require a separate model comparison.

The detector coordinates are q_z along the chosen reference axis and q_y transverse to it. For the magnetic-SANS geometry of the accompanying nanoflower manuscript, the reference axis is the applied-field direction. The code uses

$$q = \sqrt{q_z^2 + q_y^2}, \qquad \theta = \operatorname{atan2}(q_y, q_z), \qquad (q_z, q_y) = q(\cos\theta, \sin\theta). \tag{S1}$$

Angles are understood modulo 2π. The analysis domain and its area are

$$\mathcal{A} = \{(q, \theta) : q_{\min} \le q \le q_{\max},\ 0 \le \theta < 2\pi\}, \qquad A = \pi\Delta, \qquad \Delta = q_{\max}^2 - q_{\min}^2 > 0. \tag{S2}$$

Here $q_{\min} \ge 0$; when the origin is included, its angular coordinate has no effect on continuous integrals because it has zero area.

The inner product is defined with the detector-plane area measure,

$$\langle f, g\rangle_{\mathcal{A}} = \int_{q_{\min}}^{q_{\max}} \int_{0}^{2\pi} f(q,\theta) g(q,\theta)\, \mathrm{d}\theta\, q\, \mathrm{d}q. \tag{S3}$$

This is a two-dimensional reciprocal-space measure. It differs from both an unweighted radial integral $\int I\, \mathrm{d}q$ and a three-dimensional scattering invariant involving $q^2\, \mathrm{d}q$.

B. Orthonormal expansion and mode indices

The radial coordinate and radial modes are

$$x(q) = \frac{2q^2 - q_{\max}^2 - q_{\min}^2}{\Delta}, \qquad q\, \mathrm{d}q = \frac{\Delta}{4}\, \mathrm{d}x, \tag{S4}$$

$$R_m(q) = \sqrt{\frac{2(2m+1)}{\Delta}}\, P_m[x(q)], \qquad m = 0, 1, \ldots, \tag{S5}$$

where P_m is the Legendre polynomial of degree m. Since $\int_{-1}^{1} P_m(x) P_l(x)\, \mathrm{d}x = 2\delta_{ml}/(2m+1)$,

$$\int_{q_{\min}}^{q_{\max}} R_m(q) R_l(q)\, q\, \mathrm{d}q = \delta_{ml}. \tag{S6}$$

The coordinate affine in q^2 is essential for this normalization.

To distinguish the angular basis from the block statistic T_n below, we write the angular functions as $\mathcal{T}_n$:

$$\mathcal{T}_n(\theta) = \begin{cases} (2\pi)^{-1/2}, & n = 0, \\ \pi^{-1/2} \cos(n\theta), & n > 0, \\ \pi^{-1/2} \sin(|n|\theta), & n < 0. \end{cases} \qquad \int_0^{2\pi} \mathcal{T}_n \mathcal{T}_{n'}\, \mathrm{d}\theta = \delta_{nn'}. \tag{S7}$$

Thus the signed index n encodes the trigonometric branch; it is not a negative angular frequency of a complex Fourier expansion. The product modes and fitted intensity are

$$\Phi_{nm}(q,\theta) = R_m(q) \mathcal{T}_n(\theta), \qquad \widehat{I}(q,\theta) = \sum_{(n,m)\in\mathcal{K}} \widehat{c}_{nm} \Phi_{nm}(q,\theta), \tag{S8}$$

with $\langle \Phi_{nm}, \Phi_{n'm'} \rangle_{\mathcal{A}} = \delta_{nn'}\delta_{mm'}$. By default, the standard analyzer uses the full rectangle $|n| \le n_{\max}$ and $0 \le m \le m_{\max}$, containing $K = (2n_{\max}+1)(m_{\max}+1)$ product modes. With `even_only=True`, the fit uses the subset with even $|n|$ at every radial order, so that $\widehat{I}(q,\theta+\pi) = \widehat{I}(q,\theta)$. The lower-level fitting functions accept any explicitly selected set $\mathcal{K}$. Omitted coefficients are understood as zero in subsequent sums.

For a continuous, unweighted orthogonal projection of a square-integrable function, the coefficients would be

$$c_{nm}^{\mathrm{proj}} = \langle I, \Phi_{nm} \rangle_{\mathcal{A}}. \tag{S9}$$

The implemented fit instead estimates them jointly from the available detector samples. Masks, nonuniform sampling, experimental uncertainties, and finite pixels generally destroy orthogonality of the sampled columns. Consequently, Eq. (S9) is not used as a shortcut for fitting masked or uncertainty-weighted detector data.

C. Sample selection, linear fitting, and covariance

Let $p = 1, \ldots, N$ enumerate the admitted detector samples and j enumerate the selected pairs (n, m). In the default point-evaluation approximation, $X_{pj} = \Phi_j(q_p, \theta_p)$ and $y_p = I_p$. Selection requires finite coordinates and intensity, and a sample center in the chosen annulus. If uncertainties σ_p or sample weights w_p are supplied, their selected values must additionally be finite and strictly positive. Finite zero and negative background-corrected intensities remain in the fit. Logarithmic color scales affect display only.

The objective is

$$Q(\boldsymbol{c}) = (\boldsymbol{y} - X\boldsymbol{c})^{\mathsf{T}} W (\boldsymbol{y} - X\boldsymbol{c}), \qquad W_{pp} = \frac{w_p}{\sigma_p^2}. \tag{S10}$$

Absent sample weights are set to unity. If uncertainties are absent, $\sigma_p = 1$ defines the numerical metric and the residuals estimate its scale. With neither input, minimizing Q is equivalent to minimizing the ordinary sample mean squared error. The solver applies linear least squares to $W^{1/2}X$ and $W^{1/2}\boldsymbol{y}$ using `numpy.linalg.lstsq`; it does not solve a nonlinear optimization problem. A rank-deficient selected design matrix is rejected. For full column rank, the formal solution is

$$\widehat{\boldsymbol{c}} = (X^{\mathsf{T}} W X)^{-1} X^{\mathsf{T}} W \boldsymbol{y}, \qquad \nu = N - K. \tag{S11}$$

The reported `mse` always equals $N^{-1}\sum_p (y_p - \widehat{y}_p)^2$, regardless of the fitting metric. The weighted sum of squares is $Q(\widehat{\boldsymbol{c}})$; the field `reduced_chi_square` equals Q/ν. The latter has a conventional reduced-χ^2 interpretation only under the corresponding noise model.

Let $L = (X^{\mathsf{T}} W X)^{-1} X^{\mathsf{T}} W$ denote the linear map from observations to coefficients. With absolute uncertainties, the supplied σ_p are the actual standard deviations of independent

pixel measurements, so $\Sigma_{\text{abs}} = \text{diag}(\sigma_p^2)$. Additional sample weights change the fitting metric, not these measurement variances. For $\nu > 0$, the implemented coefficient covariance is

$$C = \begin{cases} L\Sigma_{\text{abs}}L^{\mathsf{T}}, & \text{absolute pixel uncertainties,} \\ \gamma(X^{\mathsf{T}}WX)^{-1}, & \text{relative precision model,} \end{cases} \qquad \gamma = Q(\widehat{\boldsymbol{c}})/\nu. \tag{S12}$$

The first case is the default when experimental uncertainties are supplied and implements the diagonal-noise specialization of the general fixed-design sampling covariance

$$C_{\text{sampling}} = (X^{\mathsf{T}}WX)^{-1}X^{\mathsf{T}}W\Sigma WX(X^{\mathsf{T}}WX)^{-1}. \tag{S13}$$

For $w_p = 1$ and $\Sigma = \Sigma_{\text{abs}}$, it reduces to the usual inverse normal matrix. For fractional area weights it generally does not: the middle factor contains w_p^2/σ_p^2, rather than w_p/σ_p^2. Multiplying all geometric weights by a common positive constant leaves both the coefficient estimates and this absolute covariance unchanged. Statistical precision factors must be incorporated in the supplied σ_p themselves.

The covariance is evaluated using a reduced QR factorization $W^{1/2}X = UR$, without forming an inverse normal matrix or an $N \times N$ observation matrix. In the absolute case,

$$L\Sigma_{\text{abs}}^{1/2} = R^{-1}U^{\mathsf{T}}\,\text{diag}(\sqrt{w_p}), \qquad C = (L\Sigma_{\text{abs}}^{1/2})(L\Sigma_{\text{abs}}^{1/2})^{\mathsf{T}}.$$

The relative case uses triangular solves for $\gamma R^{-1}R^{-\mathsf{T}}$.

Relative uncertainties can explicitly request residual scaling; fits without supplied uncertainties also use this second case. It retains the working model $\Sigma = \gamma W^{-1}$, in which the combined fit weights represent relative statistical precision. It does not calibrate geometric coverage weights with otherwise unspecified pixel-noise variance. Merely rescaling residuals does not supply the missing noise model. The implementation returns an undefined covariance when $\nu \leq 0$.

Returned errors and significance thresholds are conditional on the selected noise and linear-signal model. General non-diagonal sample covariances are not accepted; the errors do not include uncertainty in the fitted center, mode selection, or reduction-induced correlations. For deterministic simulation data, residual-scaled errors describe finite-expansion discrepancies under that model, rather than experimental counting noise.

D. Parseval identities and angular powers

For any coefficient vector in the selected orthonormal basis, including a detector-fitted vector, continuous orthonormality gives the exact identity

$$\mathcal{P} = \|\widehat{I}\|_{\mathcal{A}}^2 = \int_{\mathcal{A}} \widehat{I}^2 \, \mathrm{d}^2 q = \sum_{(n,m)\in\mathcal{K}} \widehat{c}_{nm}^2. \tag{S14}$$

For the infinite complete expansion this is Parseval's identity for I; for a finite continuous projection the omitted-mode norm is the sum of the omitted squared projection coefficients. Equation (S14) is exact for the finite reconstructed function even when its coefficients were estimated from irregular or masked data. It does not equate its continuous norm to the measured weighted residual statistic or to the norm of an unknown signal in unobserved regions.

The separate angular powers are

$$\begin{aligned}
\mathcal{P}_n^c &= \sum_{m:(n,m)\in\mathcal{K}} \widehat{c}_{n,m}^2, && n = 0, 1, \ldots, \\
\mathcal{P}_n^s &= \sum_{m:(-n,m)\in\mathcal{K}} \widehat{c}_{-n,m}^2, && n = 1, 2, \ldots, \qquad \mathcal{P}_0^s = 0, \\
\mathcal{P} &= \sum_{n\geq 0} (\mathcal{P}_n^c + \mathcal{P}_n^s).
\end{aligned} \tag{S15}$$

These powers quantify squared-intensity content on the annulus. They are neither integrated scattering cross sections nor magnetic energies, and they discard coefficient signs. For example, since $\Phi_{00} = 1/\sqrt{A}$, the linear area integral is $\int_{\mathcal{A}} \widehat{I} \, \mathrm{d}^2 q = \sqrt{A}\, \widehat{c}_{00}$ when the constant mode is present. If q has units Q and intensity has units U, then R_m has units Q^{-1}, c_{nm} has units UQ, and $\mathcal{P}$ has units U^2Q^2. Comparisons of absolute modal powers therefore require consistent coordinate units, intensity normalization, and annulus limits.

For a branch block $\widehat{\boldsymbol{c}}_n^b$ and its marginal covariance block C_n^b, with $b \in \{c, s\}$, the plotted power uncertainty is computed as

$$\delta\mathcal{P}_n^b = \left[2\,\mathrm{tr}\big((C_n^b)^2\big) + 4(\widehat{\boldsymbol{c}}_n^b)^{\mathsf{T}} C_n^b \widehat{\boldsymbol{c}}_n^b\right]^{1/2}. \tag{S16}$$

This is the square root of the Gaussian quadratic-form variance evaluated with the fitted coefficient vector, including the term quadratic in covariance. This substitution gives a plug-in uncertainty estimate rather than an exact sampling standard deviation with a known true

coefficient vector. Powers are not noise-bias corrected; squaring noisy coefficients produces positive power even for a zero true block. In the standard overview both angular power panels use

$$M = \max_{n,b} \mathcal{P}_n^b, \qquad p_n^b = \frac{\mathcal{P}_n^b}{M}, \qquad \delta p_n^b = \frac{\delta \mathcal{P}_n^b}{M}.$$

The same M is used for both branches and is held fixed in error propagation. The displayed values are not fractions of $\mathcal{P}$. The power-axis label $P_n^2/P_{\max}^2$ in the current implementation denotes $\mathcal{P}_n^b/M$; it does not square that power ratio again. On the logarithmic axes, lower error bars are shortened where necessary to remain above the positive plotting floor; the returned uncertainties retain Eq. (S16). The separate (n, m) power maps instead display $\widehat{c}_{nm}^2 / \max_j \widehat{c}_j^2$, using the largest individual product-mode power as their common scale.

E. Radial coefficient spectra associated with angular modes

Keeping each angular order separate while summing its radial modes gives the spectra returned by `angular_radial_spectra`:

$$I_n^c(q) = \sum_m \widehat{c}_{n,m} R_m(q), \quad n \geq 0, \qquad I_n^s(q) = \sum_m \widehat{c}_{-n,m} R_m(q), \quad n \geq 1. \tag{S17}$$

The stored sine spectrum at order zero is identically zero. These are signed coefficient functions with intensity units, and the reconstruction is

$$\widehat{I}(q,\theta) = \frac{I_0^c(q)}{\sqrt{2\pi}} + \frac{1}{\sqrt{\pi}} \sum_{n \geq 1} \left[I_n^c(q) \cos(n\theta) + I_n^s(q) \sin(n\theta) \right]. \tag{S18}$$

Writing the unnormalized real Fourier series as $\widehat{I} = a_0 + \sum_{n\geq 1}[a_n \cos(n\theta) + b_n \sin(n\theta)]$, the Fourier amplitudes and azimuthal average are

$$a_0(q) = \langle \widehat{I}(q,\theta) \rangle_\theta = \frac{I_0^c(q)}{\sqrt{2\pi}}, \qquad a_n(q) = \frac{I_n^c(q)}{\sqrt{\pi}}, \qquad b_n(q) = \frac{I_n^s(q)}{\sqrt{\pi}} \quad (n \geq 1). \tag{S19}$$

Thus I_0^c itself is not the azimuthally averaged intensity. The spectra retain radial sign changes and relative harmonic amplitudes that cannot be recovered from angular powers alone. Radial orthonormality connects the two descriptions exactly:

$$\int_{q_{\min}}^{q_{\max}} [I_n^b(q)]^2 q \, \mathrm{d}q = \mathcal{P}_n^b. \tag{S20}$$

In the unnormalized Fourier convention just specified, the same identities read

$$\mathcal{P}_0^c = 2\pi \int_{q_{\min}}^{q_{\max}} a_0(q)^2 q \, \mathrm{d}q,$$
$$\mathcal{P}_n^c = \pi \int_{q_{\min}}^{q_{\max}} a_n(q)^2 q \, \mathrm{d}q, \qquad \mathcal{P}_n^s = \pi \int_{q_{\min}}^{q_{\max}} b_n(q)^2 q \, \mathrm{d}q \quad (n \geq 1).$$

These factors are already included in the orthonormal coefficient convention of Eq. (S15).

Let $\boldsymbol{R}_n^b(q)$ contain the radial modes present in the selected branch block. Covariance propagation gives

$$\mathrm{Var}[I_n^b(q)] = [\boldsymbol{R}_n^b(q)]^{\mathsf{T}} C_n^b \boldsymbol{R}_n^b(q), \qquad \mathrm{Cov}[I_n^b(q), I_n^b(q')] = [\boldsymbol{R}_n^b(q)]^{\mathsf{T}} C_n^b \boldsymbol{R}_n^b(q'). \tag{S21}$$

The standard analyzer evaluates the spectra at 300 equally spaced q values and displays pointwise one-standard-deviation bands from the first expression. This grid samples the fitted function; it introduces no new independent observations. Simultaneous confidence bands are not calculated.

F. Angular block statistics T_n

Angular powers measure amplitude, whereas `angular_block_tests` tests whether an entire angular contribution is distinguishable from zero under the fit covariance model. For each branch and order the null is $H_{0,n}^b : \boldsymbol{c}_n^b = \mathbf{0}$, with every other fitted coefficient left free. The statistic and block rank are

$$T_n^b = (\widehat{\boldsymbol{c}}_n^b)^{\mathsf{T}} (C_n^b)_\tau^+ \widehat{\boldsymbol{c}}_n^b, \qquad r_n^b = \#\{j : s_j > \tau_n^b\}, \tag{S22}$$

where r_n^b is the numerical rank. For a covariance block of dimension k_n^b, a single singular-value decomposition determines both this rank and the numerical pseudoinverse:

$$C_n^b = U \, \mathrm{diag}(s_j) V^{\mathsf{T}}, \qquad \tau_n^b = k_n^b \epsilon_{\mathrm{mach}} s_{\max},$$
$$(C_n^b)_\tau^+ = V \, \mathrm{diag}(d_j) U^{\mathsf{T}}, \qquad d_j = \begin{cases} s_j^{-1}, & s_j > \tau_n^b, \\ 0, & s_j \leq \tau_n^b. \end{cases}$$

Here ϵ_{mach} is the machine precision of the double-precision covariance array. The same retained-singular-value mask is used for the rank and pseudoinverse; values exactly at the

cutoff are discarded. The latter is the Moore–Penrose pseudoinverse of the truncated covariance block and reduces to the ordinary inverse when all directions are retained. All correlations between the fitted radial coefficients enter the covariance block; individual squared coefficient-to-error ratios cannot in general replace Eq. (S22). The quantities T_n^b are dimensionless test statistics, not expansion coefficients or Parseval powers.

With independent Gaussian errors of known absolute scale and the propagated sampling covariance in Eq. (S12), the null reference distribution is $\chi^2_{r_n^b}$, including for additional geometric fit weights. When the common variance scale is estimated from the same fit residuals, the reference is $T_n^b/r_n^b \sim F_{r_n^b,\nu}$ under the relative precision model $\Sigma = \gamma W^{-1}$. This F reference is not calibrated for arbitrary geometric weights with a different noise model. Thus the plotted critical value at confidence β is

$$T^b_{n,\mathrm{crit}} = \begin{cases} \chi^2_{r_n^b;\,\beta}, & \text{absolute covariance scale,} \\ r_n^b F_{r_n^b,\nu;\,\beta}, & \text{residual-estimated covariance scale.} \end{cases} \tag{S23}$$

The default is $\beta = 0.95$; upper-tail probabilities are also returned. The code plots T_n^b itself against Eq. (S23). In the residual-scaled case it evaluates the upper tail of the $F_{r_n^b,\nu}$ distribution at T_n^b/r_n^b; the plotted statistic is not divided by r_n^b. The standard analyzer selects this case when `covariance_scaled_by_residual` is true and passes the fit's residual degrees of freedom to the block-test function. Missing blocks have rank zero and undefined statistics, including the nonexistent zero-order sine block. A nominal rejection concerns the specified block conditional on the other included modes, without correcting for the other blocks tested. A small amplitude can have a large T_n^b if it is precisely determined. The noise-model qualifications following Eq. (S12) apply to these tests.

For nearly singular blocks the statistic tests only the covariance directions retained by the common cutoff. The same numerical rank determines the reference distributions and thresholds above. An existing block with zero numerical rank has a zero stored statistic and undefined upper-tail probability and critical value; it provides no calibrated test.

The block-test result additionally provides an optional numerical Holm step-down correction for one joint family $\mathcal{H}$ containing all positive-rank cosine and sine blocks for this map, fixed annulus, and selected basis. The family size is $N_{\mathrm{test}} = |\mathcal{H}|$; missing and rank-zero blocks are excluded. For the full rectangle with $n_{\max} = 25$ and positive block ranks, $N_{\mathrm{test}} = 51$, independent of the radial truncation. Both branches are corrected together,

including the isotropic block, rather than defining a separate family for each panel.

Let $p_{(1)} \leq \cdots \leq p_{(N_{\mathrm{test}})}$ denote the ordered uncorrected upper-tail probabilities. The adjusted probabilities, mapped back to their original blocks after sorting, are

$$\widetilde{p}_{(j)} = \min\left\{1,\ \max_{1\leq i\leq j}\left[(N_{\mathrm{test}} - i + 1)p_{(i)}\right]\right\}, \qquad j = 1, \ldots, N_{\mathrm{test}}. \tag{S24}$$

The rejection rule is $\widetilde{p}_n^b \leq \alpha$, where $\alpha = 1 - \beta = 0.05$ by default. Unavailable probabilities in positive-rank blocks count conservatively as one in the adjustment and remain undefined without rejection in the output. Missing and rank-zero blocks are excluded.

Provided the marginal probabilities are valid, Holm controls the probability of rejecting any true null in this family by α under arbitrary dependence. It does not repair an incorrect covariance model, account for data-dependent basis or annulus selection, or control a larger family spanning several maps or analysis runs. The T_n^b statistics, numerical ranks, raw probabilities, and nominal critical values remain unchanged.

These error probabilities concern zero-coefficient null hypotheses under the specified noise and signal model. They do not quantify the frequency of experimental artifacts or physical misidentifications. A centering or detector artifact can produce a genuinely nonzero fitted block and hence a small model-based p-value. Calibration simulations check the implementation for their generating models; they do not establish that those assumptions hold for the measured data.

The standard overview does not highlight Holm decisions. Its dashed T_n^b curves retain the explicitly nominal, model-based thresholds of Eq. (S23). The numerical Holm properties remain available separately; they do not determine a physical identification criterion, modify the modal powers and spectra, or select and refit a reduced model.

G. Parity, mirror symmetry, and signal centering

Point reflection in the detector plane is $\theta \mapsto \theta + \pi$. Both trigonometric branches obey $\mathcal{T}_n(\theta + \pi) = (-1)^{|n|}\mathcal{T}_n(\theta)$. An exactly parity-even pattern therefore has only even angular orders. The package reports

$$f_{\mathrm{odd}} = \frac{\sum_{|n|\,\mathrm{odd},m} \widehat{c}_{nm}^2}{\mathcal{P}}, \qquad f_{\mathrm{even}} = 1 - f_{\mathrm{odd}}, \tag{S25}$$

provided $\mathcal{P} > 0$. For the fitted expansion, Parseval gives

$$D_{\mathrm{par}}^2 = \frac{1}{A}\int_{\mathcal{A}} [\widehat{I}(q,\theta) - \widehat{I}(q,\theta+\pi)]^2 \,\mathrm{d}^2 q = \frac{4\mathcal{P} f_{\mathrm{odd}}}{A}, \tag{S26}$$

$$d_{\mathrm{par}} = \frac{\|\widehat{I}(q,\theta) - \widehat{I}(q,\theta+\pi)\|_{\mathcal{A}}}{\|\widehat{I}\|_{\mathcal{A}}} = 2\sqrt{f_{\mathrm{odd}}}. \tag{S27}$$

The first measure has intensity-squared units; the second is a scale-free relative RMS mismatch and contains the same information as f_{odd}. These are diagnostics of the fitted function. With `even_only=True`, the odd power and parity mismatch vanish by construction. Mirror symmetry about the q_z axis is $\theta \mapsto -\theta$ and forbids all sine coefficients; it is an additional assumption independent of parity.

For an observable expected to be parity even, a broad background of odd angular powers can serve as an empirical reference for measurement noise, asymmetries, and processing artifacts. With the same annulus and power normalization, an even power at or below that level is not clearly resolved above this empirical background. This is a descriptive robustness criterion, not a calibrated probability threshold: mode variances, radial profiles, and sensitivities can differ. Physical odd contributions must first be excluded before interpreting odd powers as a background.

For a cross section known to be parity even, an incorrect origin can create odd harmonics. The optional centering step estimates a common displacement of the reciprocal-space origin from antipodal intensity differences. For a two-dimensional affine detector grid, the fitted coordinate map is

$$\begin{pmatrix} q_z \\ q_y \end{pmatrix} = B \begin{pmatrix} r \\ s \end{pmatrix} + \boldsymbol{b}, \qquad \boldsymbol{\delta} = B \begin{pmatrix} \delta_r \\ \delta_s \end{pmatrix}, \qquad \boldsymbol{q}_p' = \boldsymbol{q}_p - \boldsymbol{\delta}, \tag{S28}$$

where r, s are row and column indices. The antipodal partner in the original coordinates is $2\boldsymbol{\delta} - \boldsymbol{q}_p$. Bilinear interpolation gives $\widetilde{I}(2\boldsymbol{\delta} - \boldsymbol{q}_p) = \sum_j a_j I_j$ and conditional variance $\widetilde{\sigma}_p^2 = \sum_j a_j^2 \sigma_j^2$. All neighbors with nonzero interpolation weight must be valid. The minimized objective is

$$Q_{\mathrm{cen}}(\boldsymbol{\delta}) = \sum_{p \in \mathcal{H}(\boldsymbol{\delta})} \frac{[I_p - \widetilde{I}(2\boldsymbol{\delta} - \boldsymbol{q}_p)]^2}{\sigma_p^2 + \widetilde{\sigma}_p^2}. \tag{S29}$$

Here $\mathcal{H}(\boldsymbol{\delta})$ selects valid partners within the corrected annulus and one representative half-plane, avoiding explicit antipodal double counting. The admissible set can change with the trial shift. The implementation minimizes the unnormalized sum, records the before/after

objectives and pair counts, and uses a bounded Powell search. The standard analyzer bounds each detector-index displacement by two pixels by default; the standalone grid refiner defaults to one pixel. Multiple images can be passed to the standalone refiner to minimize the sum of their objectives for one common shift.

For a one-dimensional point list, the center is estimated on a regular 129×129 auxiliary grid spanning $[-q_{\max}, q_{\max}]^2$ by default. Valid source points within $1.05q_{\max}$ are used for linear intensity interpolation; centering uncertainties are assigned from the nearest measured point. After refining this grid, the displacement is applied to the original point coordinates. The final modal fit retains the original intensities, uncertainties, and point sampling. In either centering path, the selected annulus is reevaluated after the shift. Centering requires supplied uncertainties and is disabled by default in the standard analyzer. Geometry-dependent externally supplied weights must be recomputed for the shifted origin; internal pixel integration computes coverage after centering.

Interpolation correlations and center-estimation uncertainty are not propagated. Accordingly, Q_{cen} is a centering diagnostic rather than a calibrated goodness-of-fit statistic for the original data. A physical odd contribution must not be absorbed into a parity-based center. In particular, the polarization-averaged spin-flip cross section and an individual spin-flip channel need not have the same parity because an individual channel can contain a chiral contribution. An odd-power fraction alone therefore does not identify a reduction or centering error.

H. Finite pixels, rebinning, and explicit preprocessing

For an affine detector grid, optional subpixel quadrature estimates the annular coverage of a pixel P_p and the admitted-area average of each mode:

$$f_p = \frac{|P_p \cap \mathcal{A}|}{|P_p|} \simeq \frac{N_{p,\mathrm{in}}}{s_{\mathrm{sub}}^2}, \qquad \overline{X}_{pj} \simeq \frac{1}{N_{p,\mathrm{in}}} \sum_{\ell \in P_p \cap \mathcal{A}} \Phi_j(q_\ell, \theta_\ell). \tag{S30}$$

Subpixel nodes lie at the centers of a regular $s_{\mathrm{sub}} \times s_{\mathrm{sub}}$ subdivision. The affine map accounts for pixel shape and orientation. There are two distinct uses: `fractional_annulus_weights` supplies f_p as a fit weight, while `pixel_integrated_annulus_design` also replaces the design row by $\overline{X}_{pj}$. The latter admits any pixel with $f_p > 0$ and multiplies an existing sample weight by f_p. The default standard analyzer instead uses center-point selection

and evaluation; supplying only external fractional weights retains that center-point annulus selection. One must therefore enable internal pixel integration to include pixels whose centers lie outside the annulus but whose areas overlap it. Supplying the same coverage externally as well would apply it twice.

Pixel averaging refers to the portion inside the analysis annulus; the observed pixel intensity itself is not subdivided. This is an approximation for boundary pixels that represent full-pixel averages. Pixel integration also does not model wavelength spread or instrumental q resolution. With absolute errors, the supplied pixel variances remain unchanged by coverage; Eq. (S12) propagates them using the integrated design and the effective fit weights. For a point list without supplied pixel geometry, these helpers reduce to binary annulus selection and point evaluation. The integrated design is assembled in chunks to bound temporary storage.

The optional `cartesian_bin_average` routine converts point samples into equal-area Cartesian cells by

$$I_P = \frac{\sum_{\ell \in P} v_\ell I_\ell}{\sum_{\ell \in P} v_\ell}. \tag{S31}$$

For a source grid uniform in (q, θ), $v_\ell = q_\ell$ supplies the Jacobian of $q\,\mathrm{d}q\,\mathrm{d}\theta$; general nonuniform grids require their actual quadrature weights. A duplicated angular endpoint at 2π should be removed before averaging, as in the documented NuMagSANS data preparation. Empty cells remain undefined. The routine returns occupancies and source-weight sums, but does not propagate measurement uncertainties. Equal-area Cartesian samples require no additional factor of q in the subsequent area-based fit.

Further preprocessing functions are explicit optional operations. `symmetry_diagnostic` compares original pixels to bilinearly interpolated parity or mirror partners using residuals standardized by $\sqrt{\sigma_p^2 + \widetilde{\sigma}_p^2}$. Its summed statistic uses one representative half-plane and is conditional on interpolation assumptions. `symmetry_fold` averages a complete symmetry orbit $\mathcal{O}_p$ by inverse variance,

$$I_p^{\mathrm{fold}} = \frac{\sum_{j \in \mathcal{O}_p} I_j/\sigma_j^2}{\sum_{j \in \mathcal{O}_p} 1/\sigma_j^2}, \qquad (\sigma_p^{\mathrm{fold}})^2 = \left(\sum_{j \in \mathcal{O}_p} 1/\sigma_j^2 \right)^{-1}. \tag{S32}$$

Incomplete orbits are marked invalid. Parity folding imposes even orders; adding mirror folding also imposes the absence of sine contributions. The returned per-pixel errors omit correlations from interpolation and shared orbit members. Folding is intended for visualiza-

tion and sensitivity checks; symmetry-constrained fits to the original samples avoid creating these duplicated observations. Folding is never automatic.

`mask_symmetry_diagnostic` reports a weight-centroid displacement normalized by the largest admitted radius, point-reflection residuals of the mask, and the number of partially covered pixels. `rotate_coordinates` changes the angular origin according to $\theta' = \theta - \varphi$ without changing intensities. `rescale_intensity` divides intensities and uncertainties by a common positive number. `standardized_residual_outliers` flags $|I_p - \widehat{I}_p|/\sigma_p > 5$ by default and does not delete samples. These functions do not implement experimental background, efficiency, or polarization-leakage corrections; those belong to the preceding reduction.

I. Reduced reconstructions and truncation stability

The `power_model_selector` ranks individual product modes by $\widehat{c}_j^2$. For a requested power fraction $0 < \eta \leq 1$, it retains the shortest descending-power list $\mathcal{K}_\eta$ satisfying

$$\sum_{j \in \mathcal{K}_\eta} \widehat{c}_j^2 \geq \eta \mathcal{P}, \qquad \|\widehat{I} - \widehat{I}_\eta\|_{\mathcal{A}}^2 = \sum_{j \notin \mathcal{K}_\eta} \widehat{c}_j^2 \leq (1-\eta)\mathcal{P}. \tag{S33}$$

The retained coefficients are taken from the full fit without refitting. This gives a norm-controlled compression of that fitted function, not an independent test against the measured data. The standard overview reports mode counts at $\eta = 0.95, 0.98, 0.99$. Positive and negative signed indices count as distinct angular modes. The numerical result includes the reduced reconstruction and its radial spectra at a configurable threshold with default $\eta = 0.95$. The reduced spectra and their covariance blocks retain only the selected coefficients. Their propagated uncertainties condition on the selected subset.

The optional `mode_convergence_standard_analyzer` checks rectangular truncations against the largest tested rectangle, denoted by an asterisk. It constructs one maximum-order point-evaluated design matrix and refits every candidate subset with the same supplied weights. For a candidate $\mathcal{K}$, with prediction $\widehat{\boldsymbol{y}}_\mathcal{K}$ and coefficient vector zero-padded into the reference basis, it computes

$$e_{\mathrm{pred}} = \frac{\|\widehat{\boldsymbol{y}}_\mathcal{K} - \widehat{\boldsymbol{y}}_*\|_2}{\|\widehat{\boldsymbol{y}}_*\|_2}, \qquad e_{\mathrm{coef}} = \frac{\|\widehat{\boldsymbol{c}}_\mathcal{K}^{\mathrm{pad}} - \widehat{\boldsymbol{c}}_*\|_2}{\|\widehat{\boldsymbol{c}}_*\|_2}, \tag{S34}$$

$$e_{\mathrm{MSE}} = \frac{\max(\mathrm{MSE}_\mathcal{K} - \mathrm{MSE}_*, 0)}{N^{-1}\sum_p I_p^2}. \tag{S35}$$

Denominators are protected from zero numerically. All three comparisons use unweighted norms or unweighted MSE, even if the coefficient fits are weighted. A candidate is stable if all tolerances are met; their defaults are 0.01, 0.05, and 0.001, respectively. The stable rectangle with the fewest product modes is selected, with ties resolved by the sum of orders, then angular order, then radial order. If either selected order equals its tested upper limit, stabilization within the tested limits is not declared. A finite reference comparison checks numerical stability, not predictive validity or freedom from overfitting. Centering and pixel integration are not options of the current convergence wrapper; centering can be applied explicitly before a sweep.

J. Angular-template projections

Two optional template tests examine $I(q,\theta) = A(q)\sin^2\theta$ and $I(q,\theta) = A(q)\sin^2\theta\cos^2\theta$, leaving $A(q)$ unrestricted within the fitted radial basis. Since

$$\sin^2\theta = \tfrac{1}{2}[1-\cos(2\theta)], \qquad \sin^2\theta\cos^2\theta = \tfrac{1}{8}[1-\cos(4\theta)], \tag{S36}$$

the target harmonic is $h = 2$ or $h = 4$, respectively. Both require $a_h(q) = -a_0(q)$ in conventional Fourier amplitudes, or, with the present orthonormal convention,

$$c_{h,m} = -\frac{c_{0,m}}{\sqrt{2}} \quad \text{for every } m, \tag{S37}$$

and zero coefficients in all other angular contributions. The code requires matching radial index sets in the constant and target cosine blocks. With $\boldsymbol{v} = (1, -1/\sqrt{2})^{\mathsf{T}}$ and $\boldsymbol{d}_m = (\widehat{c}_{0,m}, \widehat{c}_{h,m})^{\mathsf{T}}$, its Euclidean projection is

$$\boldsymbol{d}_m^{\mathrm{tpl}} = \frac{\boldsymbol{v}^{\mathsf{T}}\boldsymbol{d}_m}{\boldsymbol{v}^{\mathsf{T}}\boldsymbol{v}}\boldsymbol{v}. \tag{S38}$$

All other template coefficients are set to zero. The squared residual splits into the mismatch within these pairs and the power in forbidden angular modes. Dividing by $\mathcal{P}$ gives the returned pair-mismatch, forbidden-mode, and total residual fractions; the explained fraction is one minus the residual fraction. This is a Parseval-metric projection, not a covariance-weighted hypothesis test. It propagates no coefficient uncertainty and imposes no positivity constraint on $A(q)$.

K. Implementation correspondence and data conventions

The definitions above correspond to `basis.py` and `indices.py` (basis and selected modes), `fit.py` (linear fit and covariance), `power.py` (powers, parity, and power selection), `spectra.py` (radial spectra), and `block_test.py` (angular block tests), all under `src/sans2d_mode_analyzer/`. Explicit detector operations reside in `preconditioning.py`, the truncation sweep in `convergence.py`, and template projections in `mode_tests.py`. The `standard_analyzer` combines sample selection, fitting over the selected modes, and the diagnostic overview. The numerical results and spectra are also returned directly.

The ILL NeXus loader maps the stored `qx1` direction to q_z, `qy1` to q_y, and converts the stored inverse-angstrom coordinates to inverse nanometers by a default factor of ten. For the bundled nanoflower data, the local data documentation identifies the 10 and 30 mT maps as $\frac{1}{2}(I^{+-} + I^{-+})$, whereas the 100 mT and 1 T maps contain the $+-$ channel alone. Their common baseline analysis uses $0.015 \leq q \leq 0.100\,\mathrm{nm}^{-1}$, the supplied uncertainties, $n_{\max} = m_{\max} = 25$, 7×7 subpixel coverage weights, and no parity centering or folding. Their supplied pixel errors are treated as absolute, with covariance propagated through the coverage-weighted fit. The ASCII loader reads columns (q_z, q_y, I, σ) without converting units; the legacy nanoporous-Fe benchmark has an unresolved coordinate-unit caveat in its data README. Its numerical q interval must therefore not be assigned physical units without establishing that provenance.

A reproducible analysis specifies the input channel and reduction, coordinate units and angular origin, annulus, retained mode set, weights and covariance scaling, any centering or symmetry operations, the truncation and compression settings, and the test confidence and family. The modal description then provides a common representation for measured and simulated detector patterns, with physical interpretation tied to the observable and assumptions used.